

Data Sources & Metadata

Project: RapidWatch — Gulf of Mexico rapid-intensification observatory **Live site:** <https://aphilp1.github.io/rapidwatch/> **This document:** complete provenance for every geospatial layer, dataset, and live feed in the project — what it is, who produced it, where it comes from, how it is fetched, its resolution / extent / coverage, the local file it lands in, the script that builds it, and its license. Maintained by hand; update when a source changes.

Coordinate reference system (everything): WGS84 geographic, lon/lat decimal degrees (`urn:ogc:def:crs:OGC:1.3:CRS84` / EPSG:4326). No projected CRS is used; the web map applies a simple equirectangular canvas projection for rendering.

Standard analysis box (Gulf of Mexico): lat 15.0 → 32.0 °N, lon −100.0 → −74.0 °E (used for all ocean-heat subsets). Hurricane selection uses a hand-drawn Gulf polygon (see `build_gulf_hurricanes.py` › `GULF_POLY`) that includes the Bay of Campeche.

"Measured" vs "illustrative": Layers are explicitly tagged below. Best tracks, D26, GOES IR, live buoys, and the NHC cone are **real observational/reanalysis data**. The Loop Current path, eddy/gliders/saildrone scaffolding, and analytical SST/shear/current climatology are **schematic** (labeled as such in-app) and are NOT in this provenance table because they are illustrative, not sourced.

1. Master quick-reference table

#	Layer / product	Provider	Type	Local file(s)	Built by	Refresh
1	RI best tracks (4 storms)	NOAA HURDAT2 + NHC TCRs	Measured	data/storms.geojson	hand-assembled	static
2	Historical Gulf hurricanes (326)	NOAA HURDAT2	Measured	data/gulf_hurricanes.geojson · .json	build_gulf_hurricanes.py	static (annual HURDAT2 release)
3	Intensification run-ups (311)	NOAA HURDAT2 (derived)	Derived	data/gulf_intensification.geojson	build_gulf_hurricanes.py	static
4	RI climatology hotspot grid	NOAA HURDAT2 (derived)	Derived	data/ri_climatology.json	ri_climatology.py	static
5	Ocean heat D26 — 2005 storms	NOAA/NRL HYCOM GOFS 3.1	Measured (reanalysis)	data/ohc/katrina_d26.json · rita_d26.json	build_ohc_hycom.py	static
6	Ocean heat D26 — 2024 storms	Copernicus Marine (GLORYS/analysis)	Measured (analysis)	data/ohc/helene_d26.json · milton_d26.json	build_ohc_copernicus.py	static
7	Scaled D26-vs-RI study	Copernicus Marine	Derived	data/ohc/ri_d26_samples.csv · ri_d26_ahead_1p0.csv	scaled_d26_ri.py · scaled_d26_ri_ahead.py	static
8	GOES IR animation — 2024	NOAA GOES-16 (ABI) via AWS	Measured	data/overlays/{helene,milton}_ir.{mp4,webm}	make_storm_ri_animation.py	static
9	GOES IR animation — 2005	NOAA GridSat-GOES (GOES-12) via NCEI	Measured	data/overlays/{katrina,rita}_ir.{mp4,webm}	make_storm_ri_animation.py	static
10	Live buoy observations	NOAA NDBC	Measured (live)	data/ndbc/*.txt	build_live_data.py	every 30 min
11	Live NHC active storms + cone	NOAA NHC	Measured (live)	data/nhc_currentstorms.json · data/nhc/*.geojson	build_live_data.py	every 30 min
12	Coastline / admin basemap	Natural Earth	Reference	data/geo/ne_50m_*.geojson	static download	static

2. Detailed source entries

1 — Rapid-intensification best tracks (the 4 headline storms)

- **What:** Synoptic best-track fixes (position, max wind, MSLP, category, RI flag, landfall flag) for Katrina (2005), Rita (2005), Helene (2024), Milton (2024). 123 point/line features.
- **Provider / source:** NOAA HURDAT2 for the 2005 pair; official **NHC Tropical Cyclone Reports** for the 2024 pair. (Recorded in the file's own `metadata.source`.)
- **RI definition:** +30 kt sustained-wind increase within 24 h on synoptic fixes (NHC operational threshold); detected from the wind series, not hand-placed.
- **Local file:** `data/storms.geojson` · **CRS:** WGS84 · **Cadence:** static.

2 — Historical Gulf hurricane track library (326 storms, 1851–2025)

- **What:** Full LineString track + lifetime-peak and in-Gulf-peak wind/category/ pressure for every Atlantic hurricane that reached HU status **and** entered the Gulf-of-Mexico polygon. 12 reached Cat 5 in-Gulf.
- **Provider / source:** NOAA HURDAT2 Atlantic best-track database, 1851–present.
- Download (once): <https://www.nhc.noaa.gov/data/hurdat/> → file `hurdat2-1851-2025-02272026.txt` (the 27 Feb 2026 release, 2004 systems).
- Stored locally as `data/hurdat2-atlantic.txt` (7.1 MB).
- **Local files:** `data/gulf_hurricanes.geojson` (tracks) · `data/gulf_hurricanes.json` (summary) · **Built by:** `build_gulf_hurricanes.py`.
- **License:** U.S. Government work, public domain. Cite NOAA/NHC; Landsea & Franklin (2013).

3 — Intensification "run-up" segments (311)

- **What:** The contiguous strengthening stretch culminating at each storm's lifetime max wind; flagged rapid (≥ 30 kt/24 h) vs gradual. 190/311 (~61%) qualified as RI.
- **Source:** derived from HURDAT2 (#2) by `build_gulf_hurricanes.py`.
- **Local file:** `data/gulf_intensification.geojson`.

4 — RI climatology hotspot grid

- **What:** Gridded conditional RI-onset propensity over the Gulf — Gaussian KDE of RI-onset start points, exposure-normalized by track traffic. 0.25° grid, 0.75° KDE bandwidth. Yields the core box (Bay of Campeche / SW Loop Current, $\sim 3.1 \times$ domain avg) and a watch box. 3744 cells.
- **Source:** derived from HURDAT2 (#2) by `ri_climatology.py`.
- **Local file:** `data/ri_climatology.json`.
- **Caveat:** Pre-1944 / pre-satellite undercount and growing observing density bias early-era counts — documented in the research memo.

5 — Ocean heat content / D26 — 2005 storms (Katrina, Rita)

- **What:** D26 = depth of the 26°C isotherm (m), computed at every grid point from 3-D ocean temperature profiles over the Gulf box for the storm's RI date.
- **Provider / source:** NOAA/NRL HYCOM GOFS 3.1 reanalysis (1994–2015).
- Dataset: `GLBv0.08/expt_53.X`, variable `water_temp`.
- Access: OPeNDAP via HYCOM THREDDS — https://tds.hycom.org/thredds/dodsC/GLBv0.08/expt_53.X/data/{year}.
- Note: GLBv0.08 longitude is 0–360 (auto-detected/converted to $-180\dots180$).
- **Extent / resolution:** Gulf box ($15\text{--}32^\circ\text{N}$, $100\text{--}74^\circ\text{W}$), native 0.08° grid ($\sim 214 \times 327$), 40 depth levels, time step nearest the RI-window midpoint.

- **Local files:** `data/ohc/katrina_d26.json` · `data/ohc/rita_d26.json` (+ `manifest.json`).
- **Built by:** `build_ohc_hycom.py` . **License:** public (NRL/NOAA); cite HYCOM/GOFS 3.1.

6 — Ocean heat content / D26 — 2024 storms (Helene, Milton)

- **What:** Same D26 product for the 2024 pair (HYCOM reanalysis does not extend past Sep 2024, so a 2024-capable source was required).
- **Provider / source:** Copernicus Marine Service — `GLOBAL_ANALYSISFORECAST_PHY_001_024` .
- Dataset id: `cmems_mod_glo_phy-thetao_anfc_0.083deg_P1D-m` (daily-mean potential temperature `thetao`), product version 202406.
- Access: `copernicusmarine` Python API (`cm.login()` / `cm.subset()`); requires a free Copernicus Marine account. **Credentials live in** `~/copernicusmarine/` **and are never committed to the repo.**
- **Extent / resolution:** Gulf box (15–32 °N, 100–74 °W), 0.083° grid, depths 0–300 m, storm's RI date.
- **Local files:** `data/ohc/helene_d26.json` · `data/ohc/milton_d26.json` .
- **Built by:** `build_ohc_copernicus.py` . **License:** Copernicus Marine Service terms — free reuse with attribution ("Generated using E.U. Copernicus Marine Service Information").

7 — Scaled D26-vs-RI statistical study

- **What:** D26 sampled at best-track fixes for 76 Gulf hurricanes (698 fixes, 89 RI-onset) to test D26 as an RI predictor; plus the **ahead-of-track** refinement that samples ~1° along each storm's heading (un-waked ocean) instead of at the cold-wake center.
- **Provider / source:** Copernicus Marine GLORYS reanalysis `cmems_mod_glo_phy_my_0.083deg_P1D-m` (fixes ≤ 2021) chained with the analysis dataset `cmems_mod_glo_phy-thetao_anfc_0.083deg_P1D-m` (≥ 2022). Cached NetCDF in `data/ohc/raw/` (gitignored).
- **Local files (headline CSVs):** `data/ohc/ri_d26_samples.csv` (center sample) · `data/ohc/ri_d26_ahead_1p0.csv` (1° ahead). Sweep CSVs are regenerable via `python scaled_d26_ri_ahead.py --sweep` .
- **Built by:** `scaled_d26_ri.py` , `scaled_d26_ri_ahead.py` , `analyze_d26_ri.py` .
- **Headline result (honest):** at center, D26 adds ~0 independent skill (AUC 0.55, $p \approx 0.07$); ahead-of-track, AUC rises to 0.64 ($p \approx 1.3e-5$) with +0.01–0.02 CV-AUC independent skill — i.e. cold-wake sampling masked the signal. Reported with its caveats in the memo §5.7.

8 — GOES infrared animation — 2024 storms (Helene, Milton)

- **What:** Reconstructed RI animation from satellite clean-IR brightness temperature.
- **Provider / source:** NOAA GOES-16 (GOES-East) ABI, Level-2 Cloud & Moisture Imagery, product `ABI-L2-CMIPC` (CONUS), **Band 13** (clean longwave IR, ~10.3 μm).
- Access: **AWS Open Data** S3 bucket `noaa-goes16` , anonymous/unsigned reads (boto3 UNSIGNED). Cached to `data/goes_cache/` .
- **Cadence (native):** ~10 min (CONUS). **Local files:** `data/overlays/{helene,milton}_ir.mp4`
- `.webm` (+ frame dirs `data/{helene,milton}_frames/` , `*_overlay_frames/`).
- **Built by:** `make_storm_ri_animation.py` <storm> . **License:** public domain (NOAA); AWS NOAA Open Data Program.

9 — GOES infrared animation — 2005 storms (Katrina, Rita)

- **What:** Same animation for the GOES-12 era (pre-ABI), so a different archive is used.
- **Provider / source:** NOAA NCEI GridSat-GOES (gridded geostationary IR), satellite `goes12` , variable `ch4` (IR window ~10.7 μm).

- Access: anonymous HTTPS — `https://www.ncei.noaa.gov/data/gridsat-goes/access/goes/{YYYY}/{MM}/GridSat-GOES.goes12.{YYYY}.{MM}.{DD}.{HH}00.v01.nc`
- **Cadence (native):** hourly (regular lat/lon grid).
- **Local files:** `data/overlays/{katrina,rita}_ir.mp4` + `.webm`.
- **Built by:** `make_storm_ri_animation.py <storm>`. **License:** public domain (NOAA NCEI).

10 — Live buoy observations (NDBC)

- **What:** Real-time marine observations (wind, gust, wave height/period, pressure, air & sea temperature, dewpoint, etc.) from Gulf moored buoys.
- **Provider / source: NOAA National Data Buoy Center (NDBC)**, realtime2 feed — `https://www.ndbc.noaa.gov/data/realtime2/{station}.txt`.
- **Stations (Gulf of Mexico moored buoys):** `42001, 42002, 42036, 42039, 42040, 42055` (station coordinates & names are on each NDBC station page; `https://www.ndbc.noaa.gov/station_page.php?station={id}`).
- **Local files:** `data/ndbc/{id}.txt` (raw realtime2, parsed by the page as-is).
- **Built by:** `build_live_data.py`. **Refresh:** every 30 min via GitHub Action.
- **Why mirrored:** NDBC has no CORS headers, so the page cannot fetch it directly in a browser — the Action mirrors it to a same-origin file.

11 — Live NHC active storms & forecast cone

- **What:** Current active tropical cyclones and their 5-day forecast cone / track / points.
- **Provider / source: NOAA National Hurricane Center (NHC)** — `https://www.nhc.noaa.gov/CurrentStorms.json` (active-storm index, verbatim).
- `https://www.nhc.noaa.gov/storm_graphics/api/{stormID}_5day_{pgn,lin,pts}.geojson` (cone polygon / track line / points per active storm).
- **Local files:** `data/nhc_currentstorms.json` · `data/nhc/{ID}_5day_{pgn,lin,pts}.geojson`.
- **Built by:** `build_live_data.py`. **Refresh:** every 30 min. (Empty when no active storms.)

12 — Coastline & administrative basemap

- **What:** Gulf coastline and admin-0 boundary lines used as the vector basemap (the app uses no raster tile service — basemap is drawn from these vectors on canvas).
- **Provider / source: Natural Earth** 50 m cultural/physical vectors (`ne_50m_admin_0_boundary_lines_land` , `ne_50m_coastline`), clipped to the Gulf and simplified. Download: `https://www.naturalearthdata.com/`.
- **Local files:** `data/geo/ne_50m_admin0.geojson` · `data/geo/ne_50m_coastline.geojson`.
- **License: public domain** (Natural Earth). Attribution appreciated, not required.

3. Live-data pipeline (the only auto-updating part)

- **Driver:** `build_live_data.py` (stdlib `urllib`, User-Agent `RapidWatch/1.0` (+`https://github.com/aphilp1/rapidwatch`)).
- **Scheduler:** GitHub Action `.github/workflows/live-data.yml`, `cron: */30 * * * *` (every 30 min; GitHub may delay/coalesce under load), Python 3.12.
- **Mechanism (git-scraping):** the Action fetches NDBC + NHC, writes the files above, and commits them (`data: refresh live buoy/NHC snapshot [skip ci]`) so the GitHub Pages site serves them same-origin — **no CORS proxy** (all free proxies are

dead/unreliable).

- **Failure mode:** on a fetch error the previous good file is kept (never overwritten with a bad/empty payload).

4. Licenses & required attribution (summary)

Source	License	Attribution string
NOAA HURDAT2 / NHC TCRs / NDBC / NHC cone / GOES-16 / GridSat-GOES	U.S. Government work, public domain	"NOAA" (cite the specific product)
NOAA/NRL HYCOM GOFS 3.1	Public (NRL/NOAA)	"HYCOM GOFS 3.1 reanalysis (NRL/NOAA)"
Copernicus Marine (GLORYS / analysis)	Free reuse w/ attribution	"Generated using E.U. Copernicus Marine Service Information"
Natural Earth	Public domain	"Natural Earth"
RapidWatch code & original content	MIT License (LICENSE)	"© Alex Philp"

Third-party datasets remain the property of their providers and carry their own terms — credit the original sources when reusing the data.

5. Software & libraries (provenance of the tooling)

- **Web map:** Leaflet (<https://leafletjs.com>) + a custom HTML5 canvas renderer for the layered raster/vector stack; all in a single self-contained page (no build step).
- **Python pipeline:** `numpy`, `netCDF4`, `requests` / `urllib`, `copernicusmarine` (v2.4.1), `boto3` (UNSIGNED S3), `scikit-learn` (D26-vs-RI study), `matplotlib` (figures), plus `ffmpeg` for animation encoding.
- **Build / assembly:** `assemble_rapidwatch.py` (page assembly + checks), `make_*` scripts (figures, animations, report/README HTML).

6. Research memo & citations

The observing-recommendation memo (`NOAA_RI_observation_report.md` / `.html`) cites 20 **verified** peer-reviewed / NOAA primary sources (DOIs confirmed) — e.g. Kaplan & DeMaria (2003), Shay (2000), Scharroo (2005), Jaimes & Shay (2009), DeMaria (1996), Landsea & Franklin (2013), Yaukey (2014), Benedetto & Mercer (2020), plus Milton/Michael TCRs. **Rule of record:** citations are never invented — each is verified against the primary source before use. The D26-vs-RI finding (including its null result on independent skill) is reported honestly.

Maintained for RapidWatch. CRS throughout: WGS84 (EPSG:4326). When a source, endpoint, dataset id, or station list changes, update the corresponding row above and the relevant `build_*` script — keep this document and the code in sync.